

Association of Windkessel Model Parameters with Local and Regional Aortic Stiffness Indices

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Abstract— Windkessel model assesses the buffering function of large arteries and has the potential to describe the global characteristics of the vasculature. In this work, we investigate the association between the Windkessel model parameters and the existing, clinically significant local, global stiffness indices and blood pulse indices. A three-element Windkessel model was used to estimate the ventricular-arterial coupling, arterial compliance, and peripheral resistance from the pressure and flow waveforms at the aorta. The study was performed on 213 healthy virtual subjects (age: 44 ± 15 years). Total arterial compliance had a significant individual correlation ($R > 0.50$, $p < 0.05$) with local and regional stiffness indices whereas characteristic impedance had a correlation of ($R > 0.22$, $p < 0.05$) with stiffness indices. Total peripheral resistance, on the other hand, did not show a significant correlation with vascular stiffness. Multivariate regression gave a better independent correlation between the Windkessel model parameters and vascular indices (Adjusted $R > 0.88$, $p < 0.05$) with total peripheral resistance showing the highest correlation (Adjusted $R = 0.95$, $p < 0.05$) and total arterial compliance showing with least correlation (Adjusted $R = 0.88$, $p < 0.05$). The significance in the association of Windkessel markers with stiffness indices highlights the potential of Windkessel model parameters in being a prognostic marker for vascular ageing. The development of an instrumentation solution for the combined assessment of pressure and flow of central arteries would accelerate the applicability of Windkessel model parameters in field settings.

Keywords— arterial stiffness, arterial compliance, characteristic impedance, flow velocity, pressure, pulse wave velocity, vascular ageing, Windkessel model

I. INTRODUCTION

According to the American heart association, if the entire population achieves higher cardiovascular health, two million major cardiovascular disease (CVD) events could be averted each year [1]. High cardiovascular health can be achieved by timely interventions of lifestyle changes and medications, by early prediction of CVDs or vascular disorders. Arterial pressure and blood flow velocity are measurable physiological parameters, that can describe the cause and acts as prognostic markers of vascular health [2].

The arterial system is characterized by ventricular-arterial coupling, arterial compliance, and peripheral resistance. Modelling of the arterial system (theoretical or numerical) explains the system characteristics and their changes using measurable physiological parameters. Windkessel model is the sole approach, for accurately assessing the buffering

function of large arteries [3]. It is a simplistic, lumped, and parameter modelling approach of the arterial system. The model describes the vasculature, based on the transmural blood pressure and blood flow velocity at the source (aortic root) coupling to the ventricle (heart).

Windkessel models can be two, three, or four elements [4]. The accuracy required in obtaining system attributes influences the choice of Windkessel modelling. The two-element model (developed and popularized by Otto Frank), consisting of arterial compliance and peripheral resistance is fairly simplistic and does not provide true systemic input impedance or predict aortic pressure in the systolic phase. To address this, a three-element Windkessel model was developed, incorporating the characteristic impedance (ventricular-arterial coupling) to account for flow resistance due to the aortic valve, which helps to improve the frequency response and provides a realistic shape of blood flow and aortic pressure [5]. Furthermore, the Windkessel configurations were eventually updated by adding more elements to the model and increasing its computational complexity. The input impedance to the arterial system was determined by the total peripheral resistance for mean pressure flow at frequencies less than 0.20 times the heart rate and by total arterial compliance for frequencies less than 3 times the heart rate and characteristic impedance for frequencies greater than 3 times heart rates [6]. The three-element model encompasses all of these elements. As a result, a three-element Windkessel model with Total Peripheral Resistance (TPR) and Total Vascular Compliance (TAC) supplemented by aortic Characteristic Impedance (Z_c) closely resembles the arterial system [5].

The Windkessel model describes the global characteristics of the artery. To practically execute the model, the pressure in the aorta is measured invasively, and the flow velocity is measured with a doppler flow instrument, however simultaneous and synchronous recording of pressure and flow velocity is a practical challenge [7], [8]. The need for such instrumentation, that can perform simultaneous acquisition of pressure and flow velocity non-invasively from central arteries, can efficiently describe the Windkessel parameters. Although the need for such instrumentation for simultaneous acquisition of pressure and flow from central arteries is emphasized, this study aims to be an early verification to analyze the relationship of Windkessel model parameters with local aortic stiffness, regional stiffness

indices, and blood pulse indices that can potentially be used for cardiovascular pathophysiological diagnosis. In this study, we have investigated the i) Windkessel model parameters (global stiffness) derived from the three-element Windkessel model applied on aortic pressure and flow velocity waveforms and ii) its associations with clinically relevant (local) vascular material properties and ageing markers derived from the aortic pressure and flow velocity on virtual subjects. The subsequent sections describe the methodology, followed by results and observations accompanying insights to future works.

II. MATERIALS AND METHODS

A. Estimation of Windkessel Model Parameters

For a given input pressure ($P(t)$) and flow velocity ($U(t)$) (analogous to a voltage source ($v(t)$) and current source $i(t)$ in electrical model), the three-element Windkessel model, is represented in the hydraulic form as depicted in Fig. 1(a). The given model is characterized by its Z_c analogous to proximal series resistance, TAC, and TPR representing capacitor in parallel and distal resistor respectively. In this study, the electrical representation of the three-element Windkessel model (Fig. 1 (b)) is used to solve for the model parameters.

Applying Kirchhoff circuit laws [9] on the given electrical network, $i(t)$ is expressed as,

$$i(t) = \frac{v(t)}{R} - \frac{r(i(t))}{R} + C \left(\frac{dv(t)}{dt} - r \left(\frac{di(t)}{dt} \right) \right) \quad (1)$$

converting (1) into Laplace domain, yields,

$$I(s) = \frac{V(s)}{R} - \frac{r}{R} I(s) + C (sV(s) - r(sI(s))) \quad (2)$$

now, (2) can be re-written as a second-order transfer function of the form,

$$\frac{I(s)}{V(s)} = \frac{1 + RC_1 s}{rCs + (r + R)} \quad (3)$$

For a given input $i(t)$ and $v(t)$, (3) is solved computationally to get the coefficients of the numerator and denominator of the expected transfer function model, as

$$\frac{I(s)}{V(s)} = \frac{1 + K_1 s}{K_2 s + K_3} \quad (4)$$

where, K_1, K_2, K_3 are the coefficients of the transfer function, comparing (3) with (4), yields,

$$K_1 = RC \quad (5)$$

$$K_2 = rC \quad (6)$$

$$K_3 = r + R \quad (7)$$

Solving (5), (6), and (7), r, R , and C are estimated. Converting the electrical parameters into hydraulic form, Z_c , TPR, and TAC are estimated.

B. Estimates of the Vascular Stiffness Indices (Regional and Local) And Blood Pulse Indices

The stiffness index (β), pressure strain elasticity (E_p), arterial compliance (AC), and pulse wave velocity (PWV) are the clinically relevant indices for estimating vascular stiffness. β is given by,

$$\beta = \frac{\ln(P_s/P_d)}{\Delta D/D_d} \quad (8)$$

Where P_s and P_d are the systolic and diastolic blood pressure, ΔD is the distension and D_d is the end diastolic diameter. E_p is calculated using:

$$E_p = \frac{D_d \times \Delta P}{\Delta D} \quad (9)$$

Where ΔP is the pulse pressure. Using the systolic and diastolic diameter (D_s and D_d), the arterial compliance parameter was measured by:

$$AC = \frac{\pi(D_s^2 - D_d^2)}{4\Delta P} \quad (10)$$

Finally, PWV was measured using the Bramwell-Hill equation [10]:

$$PWV = \sqrt{\frac{D \times dP}{2\rho \times dD}} \quad (11)$$

Where D is the lumen diameter, P is the pressure measured, and ρ is the blood density.

The estimation of regional stiffness was done using carotid to femoral PWV (regional PWV) which is the recommended method for measuring arterial stiffness [10] [11]. The blood pulse indices (Left ventricular ejection time [LVET] and Cardiac Output [CO]) are obtained from the dataset. The calculation of the Augmentation Index (AIx) is discussed in the subsequent section.

C. Study Population

The virtual subjects database developed by the Hemodynamic Modelling Research Group of Kings College, London, UK, [12] was used in this study. The database was created by varying the hemodynamic parameters (cardiac, arterial, and blood properties) of a healthy subject. A subset of 213 subjects consisting of an equal proportion of Type-A,

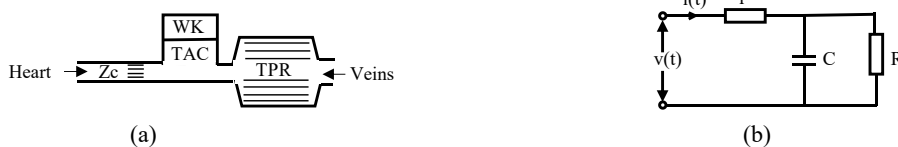


Fig. 1. The three-element Windkessel model in (a) hydraulic form and (b) electrical form.

Type-B, and Type-C waveform morphology (measured with respect to the location of peak systolic pressure and AIx as described elsewhere [13]) and spread across the age group ($n = 70$ subjects in each group with age = 44 ± 15 years) was used in this study ensuring data diversity.

D. Data Processing

Pressure (in mmHg), flow velocity (in m/s), and diameter (in mm) waveforms (single cycle per subject) at aorta for the selected subset of the database were sampled at 500 Hz. The cycles were defined from minima to minima using a cycle trimming algorithm for respective pressure and flow velocity. Baseline wandering, if any was compensated by a linear baseline estimator. Solving the Windkessel transfer function model necessitated a continuous set of pressure and flow velocity cycles for each subject. This was created by repeatedly concatenating the given cycle's minima to minima and smoothing it with a moving average filter ($M = 5$ points). The type of the waveform morphology (Type-A, Type-B, Type-C) and AIx [14] were identified from the higher derivative waveforms [15] and filtered using a Savitzky-Golay filter with 4th order polynomial and 36 side points [16].

Stiffness indices (β , Ep, AC, and PWV) from pressure and diameter waveforms for each subject, as described in equation (8)-(11), and corresponding Windkessel model parameters (Z_c , TPR, TAC) from the transfer function model are implemented by optimizing the coefficients of the continuous-time (single input single output) transfer function model in LabVIEW® platform (National Instruments, USA) with a set of model accuracy indices to ensure data fitting efficiency for each subject.

The estimated Windkessel indices were validated using a PSpice circuit simulation (TINA®, Texas Instruments) software. An electrical equivalent model of the three-element Windkessel model was made with the known pressure waveforms being the source and the model parameters being the resistive and capacitive elements.

E. Statistical Analysis

Continuous values are presented as a range and/or mean $\pm$ standard deviations. Regression analysis was used to describe the correlation, in terms of Pearson coefficient (R) and statistical significance in terms of the p -value (Students' T-test). A p -value less than 0.05 (level of significance, α) is considered statistically significant. To compare measurements between any two types, the percentage difference in group-average value was employed, and the ANOVA test was used to validate statistical significance ($p < 0.05$). The independence of association between variables was studied using multivariate regression analysis.

III. RESULTS AND DISCUSSION

A. Association of Windkessel Markers with Local Stiffness Indices

The flow estimates validated using the TINA® software had a root mean square error < 0.05 m/s. Fig. 2 illustrates the relation between Windkessel markers and stiffness indices. TAC shows a consistent correlation with all the stiffness

markers except for AIx ($R = -0.07$, $p > 0.05$). On the other hand, TPR being the peripheral resistance showed less/no correlation with the local and regional stiffness markers. Z_c has a moderate correlation with AC and LVET ($R > 0.05$, $p < 0.05$). A statistically significant correlation ($R > 0.45$, $p < 0.05$) was obtained between TAC and clinically relevant local stiffness indices (β , Ep, AC, PWV) the local PWV being the highest correlated ($R = -0.56$, $p < 0.05$). TAC is having a decreasing trendline with β ($R = -0.45$, $p < 0.05$), Ep ($R = -0.48$, $p < 0.05$), and PWV and an increasing trendline with AC ($R = 0.55$, $p < 0.05$). Compared to TAC, Z_c was less correlated with local stiffness indices ($R > 0.20$, $p < 0.05$) with AC having the highest correlation with Z_c ($R = -0.53$, $p < 0.05$). There existed a polynomial trendline between Windkessel markers and local stiffness indices. TPR was not having a significant correlation with the local stiffness indices. Table 1. shows the multivariate regression analysis detailing the independent association of Windkessel markers with stiffness. TAC shows a moderate correlation with local stiffness indices with β having the highest R -value of 0.471. The p -value (> 0.05) indicates independency of Z_c over both regional and local PWV, TAC over AC and AIx, and TPR over regional, local PWV and AIx. Therefore, both Z_c and TAC have the potential to be employed as independent stiffness markers in addition to clinical markers for a more accurate assessment of the vasculature. Z_c and TAC along with clinical indices like β , Ep, AC, and PWV and continuous monitoring of heart activity [17] may further improve the estimation of EVA.

The elastic property of the artery helps in maintaining a constant blood flow throughout the vessel. As stiffness increases, the pulsatile energy storage capacity reduces thereby increasing the pulse wave velocity. The storage capacity is analogous to the total peripheral compliance and this explains the inverse correlation of TAC with stiffness indices. As the Aorta gets stiffer, AC decreases [18]. It also attributes to a decrease in the effective aortic diameter and thereby Z_c which is inversely dependent with diameter increases. The same negative correlation between Z_c and AC was observed in the analysis. TPR depends more on the peripheral vasculature and has less effect on the aorta as seen in the lower correlation score owing to the inverse relation of resistance with vessel diameter.

B. Association of Windkessel Markers with Regional Stiffness Index

A statistically moderate significance ($R = 0.49$, $p < 0.05$) was obtained between TAC and regional PWV (carotid to femoral PWV). A polynomial decreasing trendline was observed between TAC and PWV. The correlation of PWV with Z_c was comparatively lesser ($R = 0.22$, $p < 0.05$) and TPR did not show a significant correlation ($p > 0.05$) with regional PWV. In multi regression analysis, TAC shows a significant correlation with regional PWV.

Windkessel resistance values are not correlated with regional stiffness indices [19] since it has a direct association with heart rate and systemic vascular resistance alone. TAC on the other hand has a direct inverse relationship with regional PWV [20].

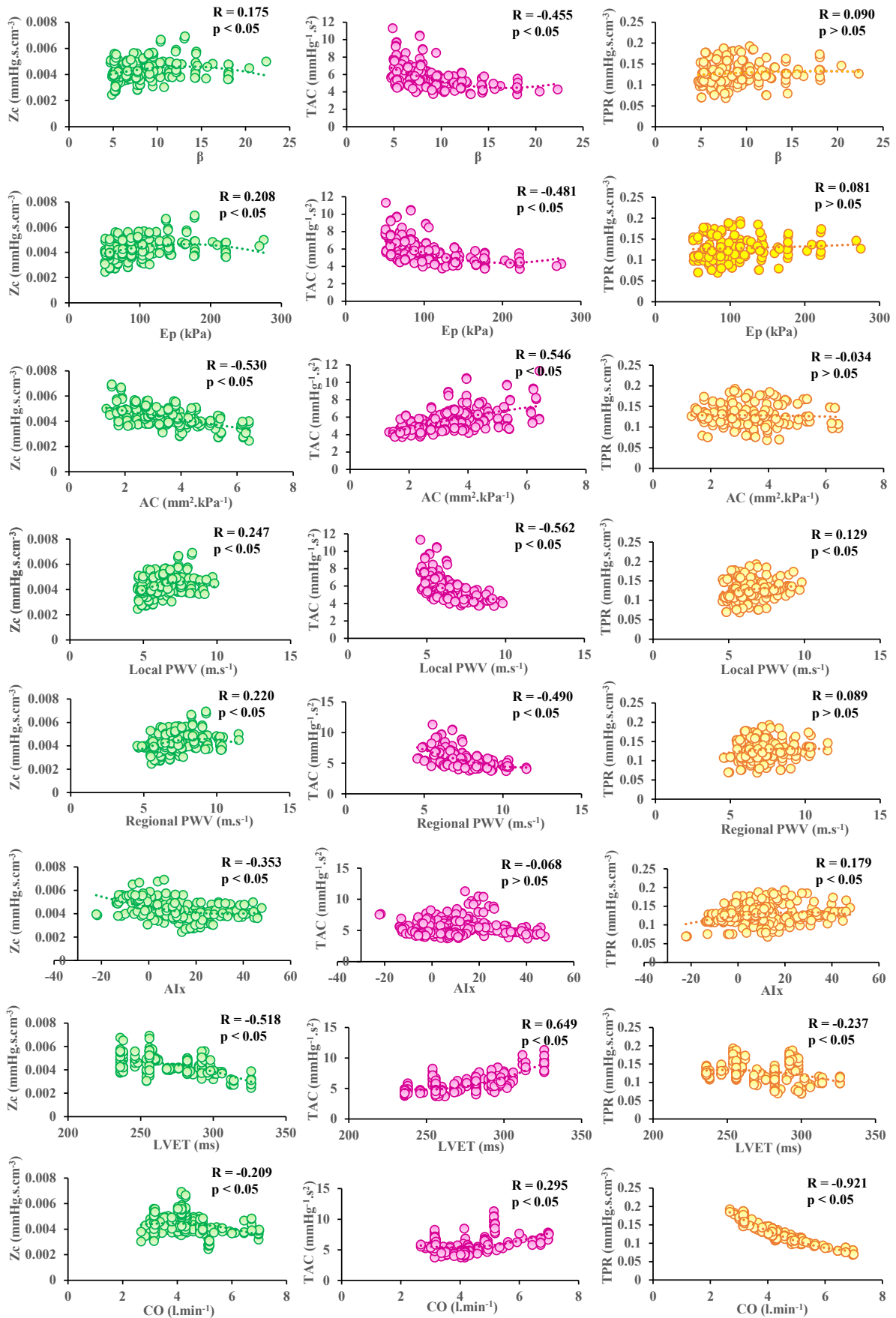


Fig. 2. Correlation between Windkessel model parameters (Zc, TAC and TPR) with stiffness indices β , Ep, AC, PWV, and blood pulse indices (AIx, LVET, CO) for 213 virtual subjects.

TABLE 1. RESULTS OF MULTIVARIATE REGRESSION ANALYSIS

Model	Dependent Variables	Coefficients	P-value	Lower 95%
Zc Multiple R: 0.938 Adjusted R: 0.936 $p < 0.05$	Beta	-0.001	0.001	-0.001
	Ep	0.001	0.001	0.001
	AC	-0.001	0.001	-0.001
	BH PWV	-0.001	0.454	-0.001
	cf_PWV	0.001	0.074	-0.001
	AIx (Aorta)	-0.001	0.001	-0.001
	LVET [ms]	0.001	0.004	0.001
	CO [l/min]	-0.001	0.001	-0.001
TAC Multiple R: 0.889 Adjusted R: 0.884 $p < 0.05$	Beta	0.471	0.001	0.291
	Ep	-0.034	0.001	-0.054
	AC	0.022	0.078	-0.003
	BH PWV	-1.277	0.001	-1.837
	cf_PWV	0.528	0.037	0.032
	AIx (Aorta)	0.006	0.463	-0.011
	LVET [ms]	0.031	0.001	0.023
	CO [l/min]	0.145	0.013	0.030
TPR Multiple R: 0.951 Adjusted R: 0.949 $p < 0.05$	Beta	-0.008	0.001	-0.011
	Ep	0.001	0.001	0.001
	AC	-0.001	0.028	-0.001
	BH PWV	0.005	0.167	-0.002
	cf_PWV	-0.004	0.212	-0.011
	AIx (Aorta)	0.001	0.871	-0.001
	LVET [ms]	0.001	0.001	0.001
	CO [l/min]	-0.029	0.001	-0.030

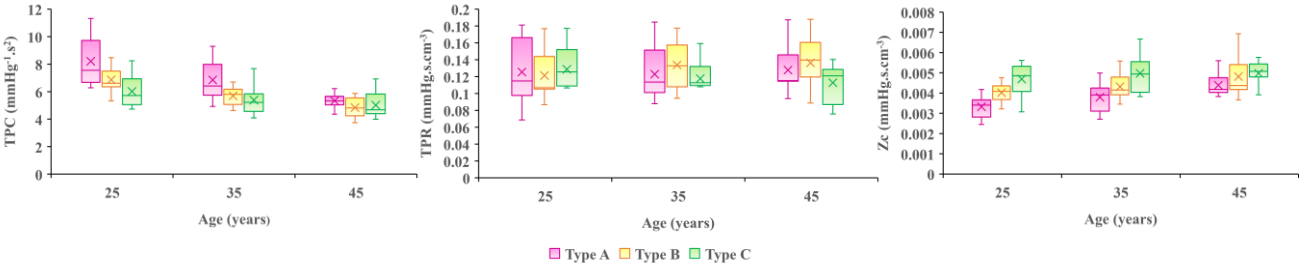


Fig. 3. Distribution of Windkessel parameters for Type-A, Type-B, and Type-C across age decades.

C. Association of Windkessel Markers with Blood Pulse Indices

Windkessel markers didn't show a significant correlation with AIx (Zc showing the maximum correlation ($R = -0.35$, $p < 0.05$)). Whereas, TAC showed a significant correlation with LVET ($R = 0.65$, $p < 0.05$) followed by Zc ($R = -0.52$, $p < 0.05$). A polynomial trendline was observed between Windkessel markers and blood pulse indices. TPR showed a comparatively lower correlation with AIx and LVET but have a strong correlation with CO ($R = -0.92$, $p < 0.05$). No significant correlation is observed for multiple regression analysis between Windkessel markers and blood pulse indices. TPR is directly obtained from CO and BP [21] and thus has less influence on stiffness indices which indeed is reflected in the correlation values. The Windkessel model site also plays a crucial role in varying the Windkessel markers. For example, TPR would have a larger significance if the model site was a small artery.

D. Distribution of Windkessel Model Parameters for Type-A, Type-B, and Type-C subjects.

The variation of Windkessel parameters on Type-A, Type-B, and Type-C subjects for the age distribution of 35 ± 8 years is given in Fig. 3. There is a significant decrease ($p < 0.05$) in the TAC values between Type-A and Type-B subjects (14.58%) and a moderate decrease between Type-B and Type-C (5.68%) whereas Zc shows an increase of 11.62% in the group average value between Type-A and Type-B and 10.41% between Type-B and Type-C which is in line with the literature [6]. There was a slight variation of TPR between types (3.85% increase between Type-A and Type-B, and 8.46% decrease between Type-B and Type-C).

Although the three-element Windkessel model cannot very well describe the high-frequency details (the inflection point and augmentation in aortic pressure) required to classify types between subjects, the results obtained show a significant relation to the pressure waveform morphology. A

smaller value of Z_c among Type-A suggests considerable reflections in the artery whereas a slightly higher value of Z_c for Type-B and Type-C implies more defused reflections. Similarly, a higher value of TAC indicates an augmented aortic pressure which is common in Type-A subjects.

E. Limitations and Future Scope

The study constraints, particularly the use of virtual subjects' data for associating the Windkessel markers with the stiffness indices, should be highlighted when interpreting reported observations. This is a pre-verification study done to analyse the variation of Windkessel markers given the blood pressure and flow velocity and its association with the (local) vascular material properties and ageing markers and requires a clinical validation/field study to be done on human subjects.

Typically, the Windkessel markers are calculated using pressure and flow waveform measured using an invasive catheter or any sequential measurement procedures. Thus, there is a lack of instrumentation for the combined assessment of arterial pressure and blood flow through the arteries. Real-time continuous pressure can be recorded by cuffless/cuff-based approaches [22] based on local PWV [23] and its incremental nature [11]. By reliable recording of continuous changes in arterial wall dynamics (diameter) using efficient signal processing algorithms [24], the local stiffness indices for a target artery can be calculated [25], [26]. Development of a multi-modal acquisition system for simultaneous and synchronous pressure, diameter and flow, estimating the local/regional stiffness and Windkessel parameters that can be used in a resource-constrained community/field setting is in progress.

IV. CONCLUSION

In this work, we have derived the Windkessel model parameters from the pressure and flow velocity values of 213 virtual subjects using the three-element Windkessel model and presented the association of these markers with local and regional stiffness and blood pulse indices. The study revealed a significant association of Windkessel markers with clinically relevant stiffness indices. Further, the multivariate regression analysis depicted an independent association of Z_c and TAC with arterial stiffness. The combined use of Windkessel markers and stiffness indices in early vascular screening needs to be further elucidated. There existed a statistically significant correlation between both the markers which points to the need for an instrumentation system that can simultaneously capture the blood pressure and blood flow velocity. The association between Windkessel model parameters and stiffness indices can be further studied on large clinical setups.

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